

STUDENT PERFORMANCE TRACKER USING MACHINE LEARNING

INTERNSHIP PROJECT REPORT

Submitted By

Name: Kovvuri Jahnavi

Intern ID: CITS943

College: Aditya University

Branch: MCA

Organization: CODTECH IT SOLUTIONS

Duration: 6 Weeks

ABSTRACT

Student Performance Tracker is a Machine Learning-based system used to analyze and predict student academic performance. It helps in understanding student progress using factors such as attendance, study hours, assignments, and previous marks.

In this project, machine learning algorithms are used to predict student performance and identify students who need improvement support.

INTRODUCTION

Student performance analysis is an important part of the education system. Manual evaluation takes time and may not always be accurate.

This project aims to build an automated system that analyzes student data and predicts academic performance using machine learning techniques.

OBJECTIVES

- To build a student performance prediction system
 - To analyze academic data of students
 - To identify factors affecting performance
 - To predict student marks using machine learning
 - To support teachers in decision-making
-

TECHNOLOGIES USED

Python
Pandas
NumPy
Matplotlib
Seaborn
Scikit-learn

METHODOLOGY

Data collection of student records
Data cleaning and preprocessing
Exploratory Data Analysis (EDA)
Feature selection (attendance, marks, study hours)
Splitting dataset into training and testing data
Model training using Linear Regression / Decision Tree Regression
Model evaluation using error metrics
Prediction of student performance

DATASET DESCRIPTION

The dataset contains student information such as attendance, internal marks, assignment scores, study hours, and final exam marks. These features are used to predict overall academic performance.

IMPLEMENTATION

The system is developed using Python and Scikit-learn. Data preprocessing is performed to clean and organize the dataset.

Machine Learning models like Linear Regression and Decision Tree Regression are trained using student data. The trained model is then used to predict student performance for new inputs.

RESULTS

The system successfully predicts student performance based on input features. It helps identify weak students and supports teachers in improving academic results.

ADVANTAGES

Easy performance tracking
Identifies weak students early
Reduces manual work
Improves academic planning
Provides data-driven insights

APPLICATIONS

Schools and colleges
Online learning platforms
Academic monitoring systems
Student counseling systems
Educational analytics dashboards

FUTURE ENHANCEMENTS

Use of deep learning models for better accuracy
Real-time performance tracking system
Web-based dashboard for teachers
Integration with LMS platforms
AI-based student guidance system

CONCLUSION

The Student Performance Tracker project demonstrates how Machine Learning can be used in education to analyze and predict student performance. It helps improve learning outcomes and supports academic decision-making.

REFERENCES

Python Documentation
Pandas Documentation
NumPy Documentation
Scikit-learn Documentation
Research papers on educational data mining

ACKNOWLEDGEMENT

I sincerely thank CODTECH IT SOLUTIONS for providing me the opportunity to complete this internship project. I also express my gratitude to Aditya University, faculty members, and mentors for their continuous support and guidance.